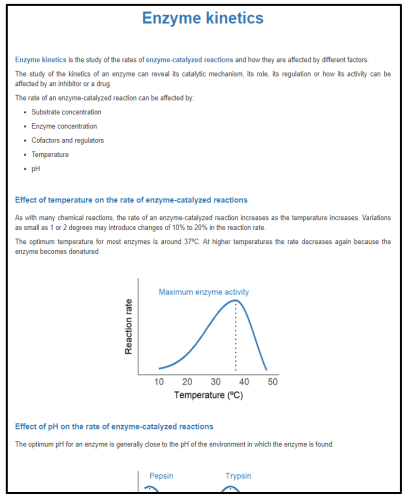


Practical enzymology

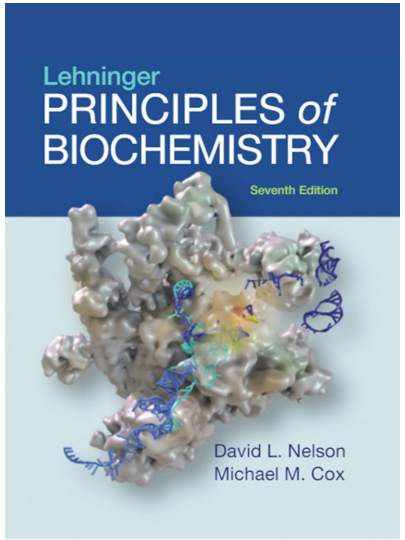
Enzyme kinetics and inhibition

Supportive material



Additional study material

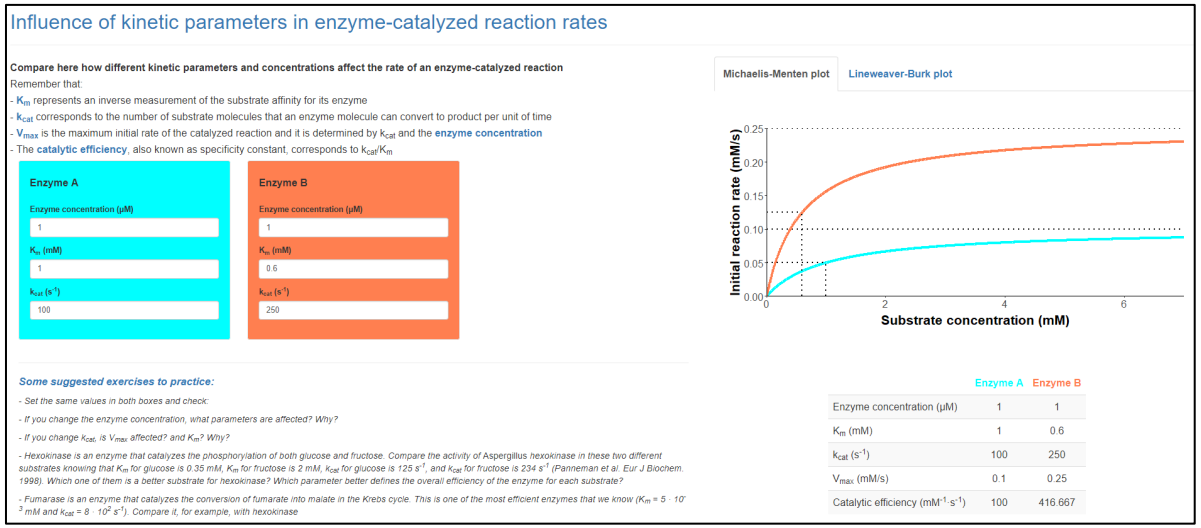
Enzyme kinetics (in Aula Virtual)



Reading

Lehninger's PoB Chapter 6:

6.3. Enzyme kinetics



Practice

Web tool:

https://rpio.shinyapps.io/Enzyme_kinetics_practice

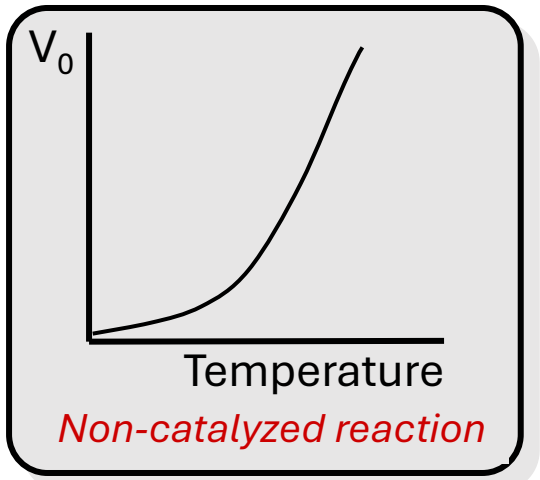
Learning goals

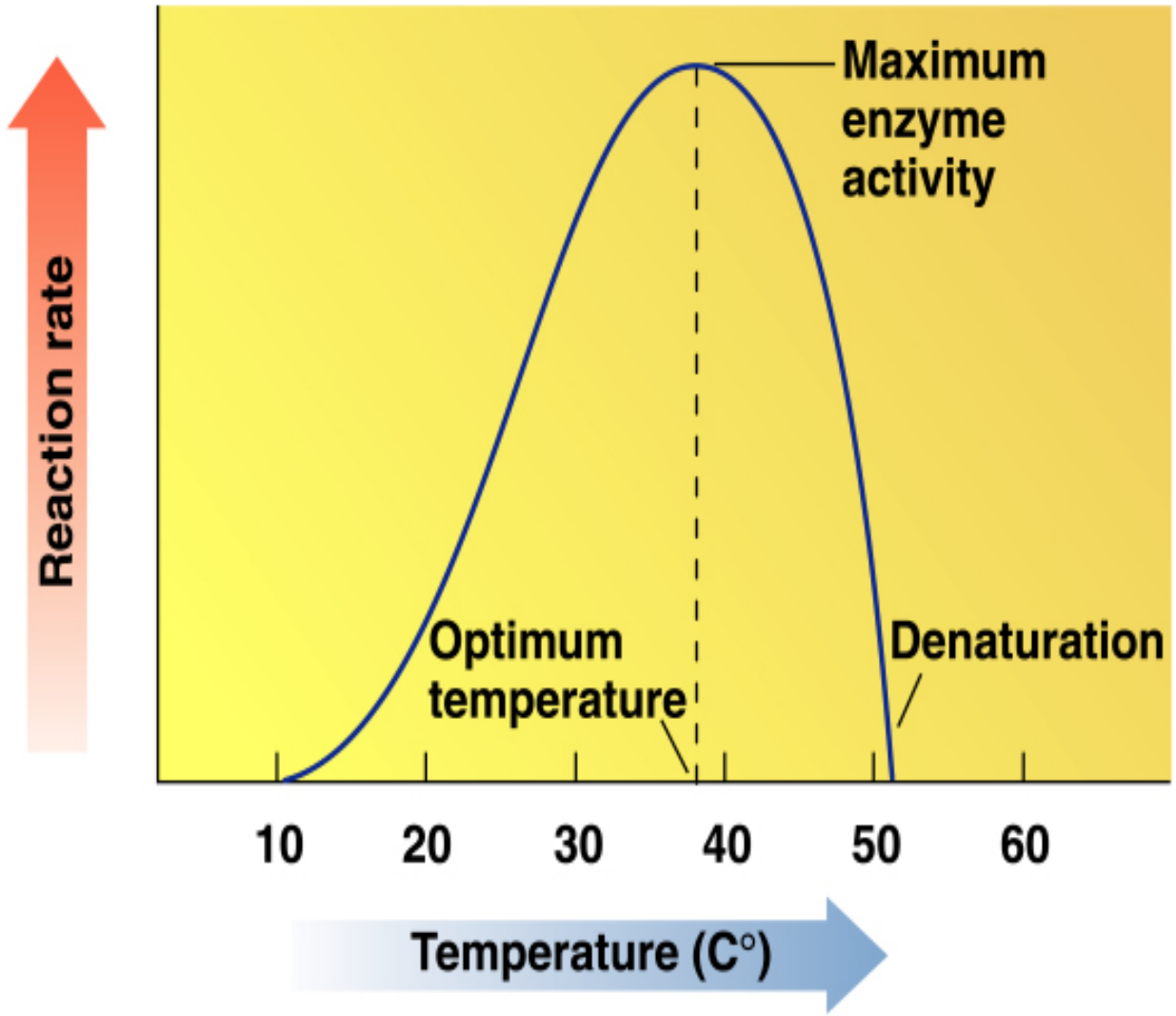
- Factors that affect enzyme activity
- Michaelis-Menten model and equation
- Kinetic parameters in enzyme-catalyzed reactions
- Graphical determination of kinetic parameters
- Graphical determination of enzyme inhibition

Enzyme kinetics

- Enzyme kinetics is the study of the rate of an enzyme-catalyzed reaction
- Studying an enzyme's kinetics can reveal its catalytic mechanism, its role, its regulation or how its activity can be affected by a drug
- The rate of an enzymatic reaction is affected by:
  - Enzyme concentration
  - Cofactors and regulators
  - Temperature
  - pH
  - Substrate concentration

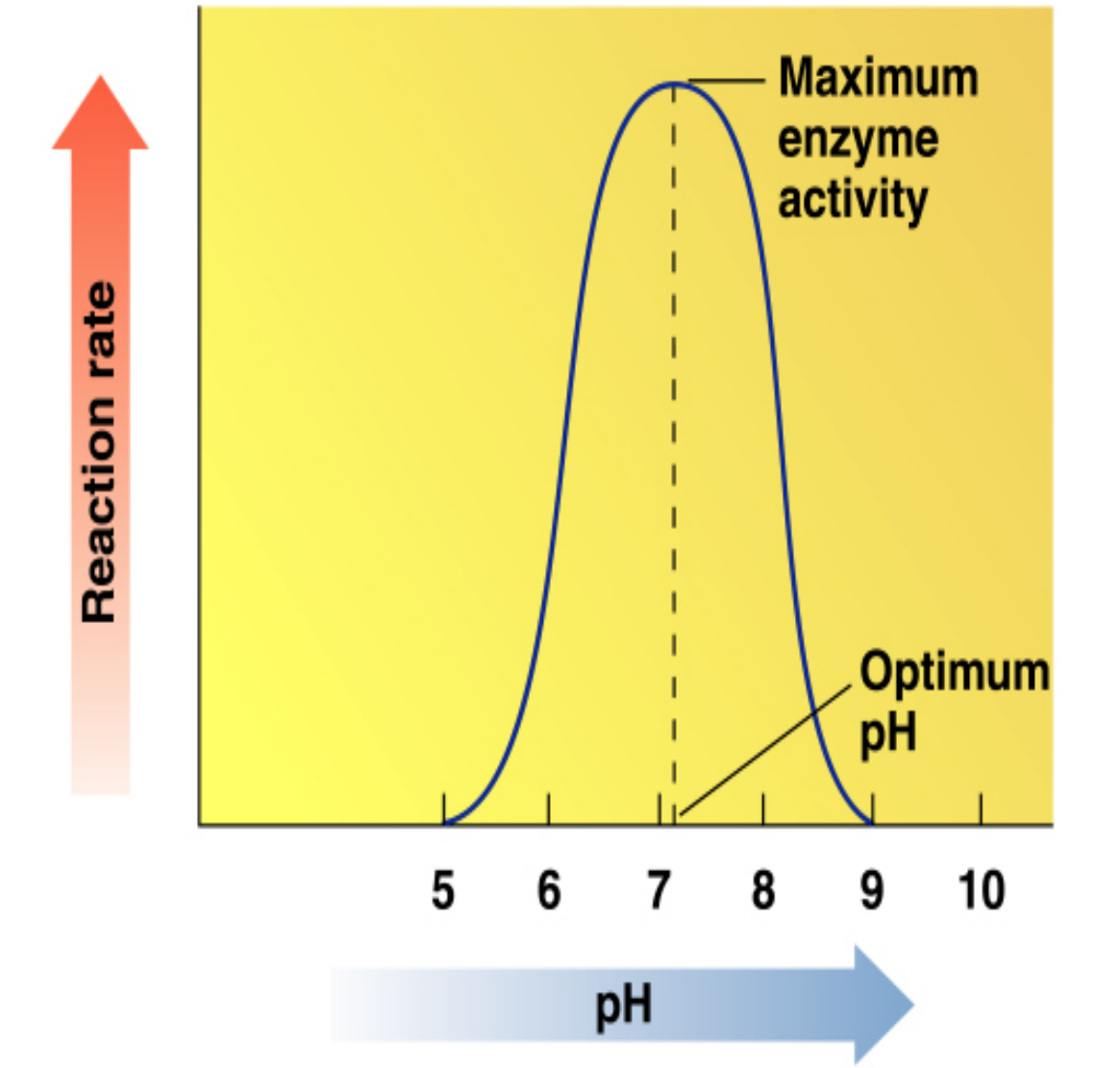
Effect of temperature





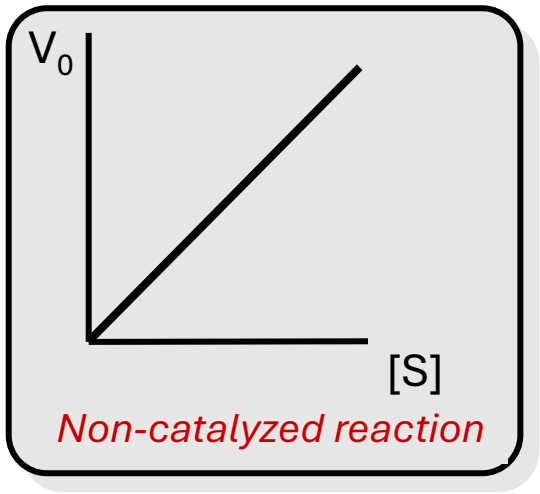
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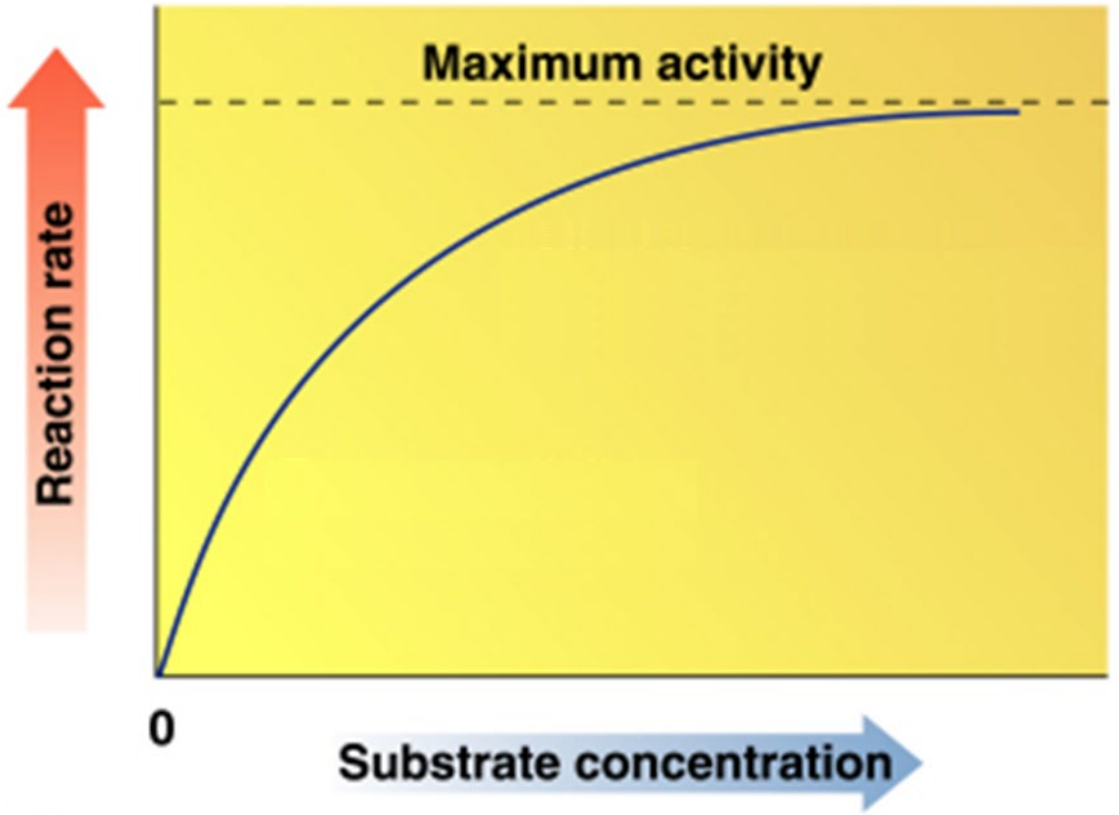
Effect of pH



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Effect of substrate concentration





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Enzymes can be saturated with substrate

Michaelis-Menten model



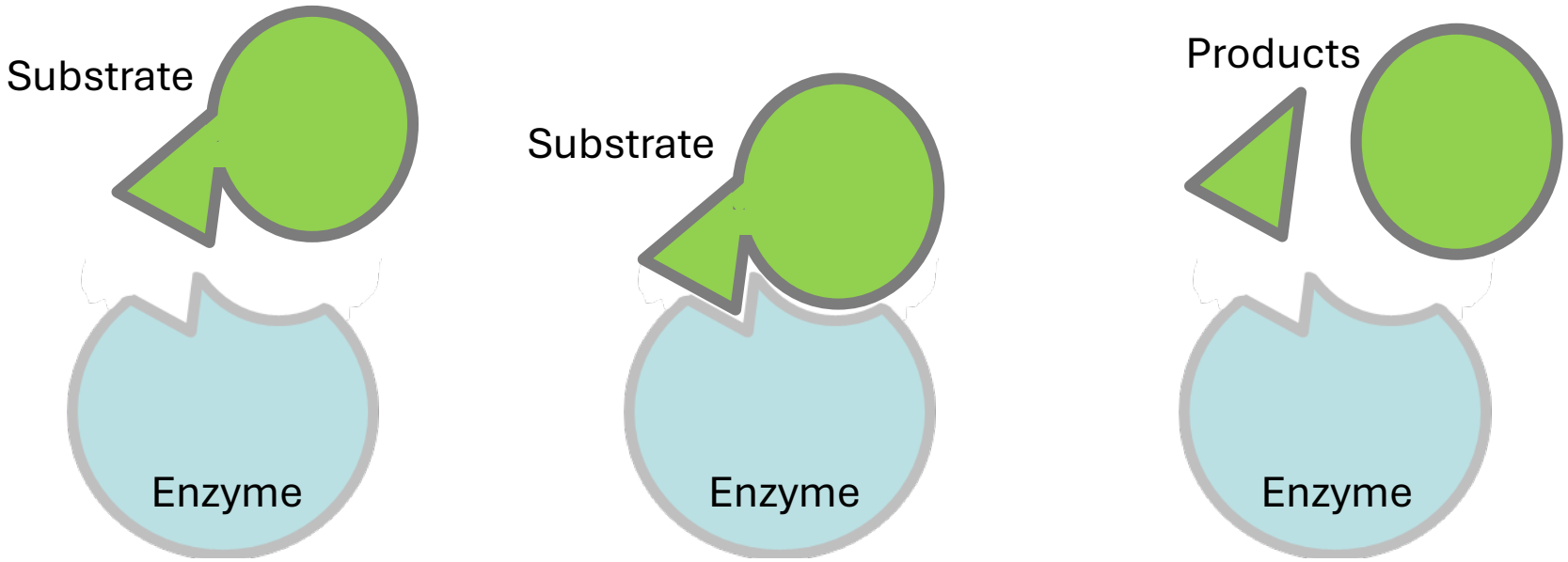
Leonor Michaelis
(1875-1949)



Maud Leonora Menten
(1879-1960)

Michaelis-Menten model

$$E + S \rightleftharpoons ES \longrightarrow E + P$$



Michaelis-Menten model

$$V_0 = \frac{V_{max}[S]}{K_m + [S]}$$

K_m may reflect the stability of the ES complex:

K_m high → weak

K_m low → strong

TABLE 6-6	K_m for Some Enzymes and Substrates	
Enzyme	**Substrate**	**K_m (mM)**
Hexokinase (brain)	ATP	0.4
	D-Glucose	0.05
	D-Fructose	1.5
Carbonic anhydrase	HCO_3^-	26
Chymotrypsin	Glycyltyrosinylglycine	108
	N-Benzoyltyrosinamide	2.5
β-Galactosidase	D-Lactose	4.0
Threonine dehydratase	L-Threonine	5.0

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Turnover number (k_{cat})

- Number of substrate molecules that an enzyme molecule can convert to product per unit of time
- Number of catalytic cycles per unit of time

$$k_{cat} = \frac{V_{max}}{[E_T]}$$

TABLE 6-7	Turnover Number, k_{cat}, of Some Enzymes	
Enzyme	**Substrate**	**k_{cat} (s^{-1})**
Catalase	H_2O_2	40,000,000
Carbonic anhydrase	HCO_3^-	400,000
Acetylcholinesterase	Acetylcholine	14,000
β-Lactamase	Benzylpenicillin	2,000
Fumarase	Fumarate	800
RecA protein (an ATPase)	ATP	0.5

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Catalytic efficiency

Neither the turnover number (k_{cat}) nor the Michaelis-Menten constant (K_m) alone are sufficient to evaluate the enzyme performance.

The best way to express the catalytic efficiency of an enzyme is to combine both parameters:

$$\text{Specificity constant} = \frac{k_{cat}}{K_m}$$

TABLE 6-8 Enzymes for Which k_{cat}/K_m Is Close to the Diffusion-Controlled Limit (10^8 to 10^9 $M^{-1}s^{-1}$)

Enzyme	Substrate	k_{cat} (s^{-1})	K_m (M)	k_{cat}/K_m ($M^{-1}s^{-1}$)
Acetylcholinesterase	Acetylcholine	1.4×10^4	9×10^{-5}	1.6×10^8
Carbonic anhydrase	CO_2	1×10^6	1.2×10^{-2}	8.3×10^7
	HCO_3^-	4×10^5	2.6×10^{-2}	1.5×10^7
Catalase	H_2O_2	4×10^7	1.1×10^0	4×10^7
Crotonase	Crotonyl-CoA	5.7×10^3	2×10^{-5}	2.8×10^8
Fumarase	Fumarate	8×10^2	5×10^{-6}	1.6×10^8
	Malate	9×10^2	2.5×10^{-5}	3.6×10^7
β-Lactamase	Benzylpenicillin	2.0×10^3	2×10^{-5}	1×10^8

Source: A. Fersht, *Structure and Mechanism in Protein Science*, p. 166, W. H. Freeman and Company, 1999.

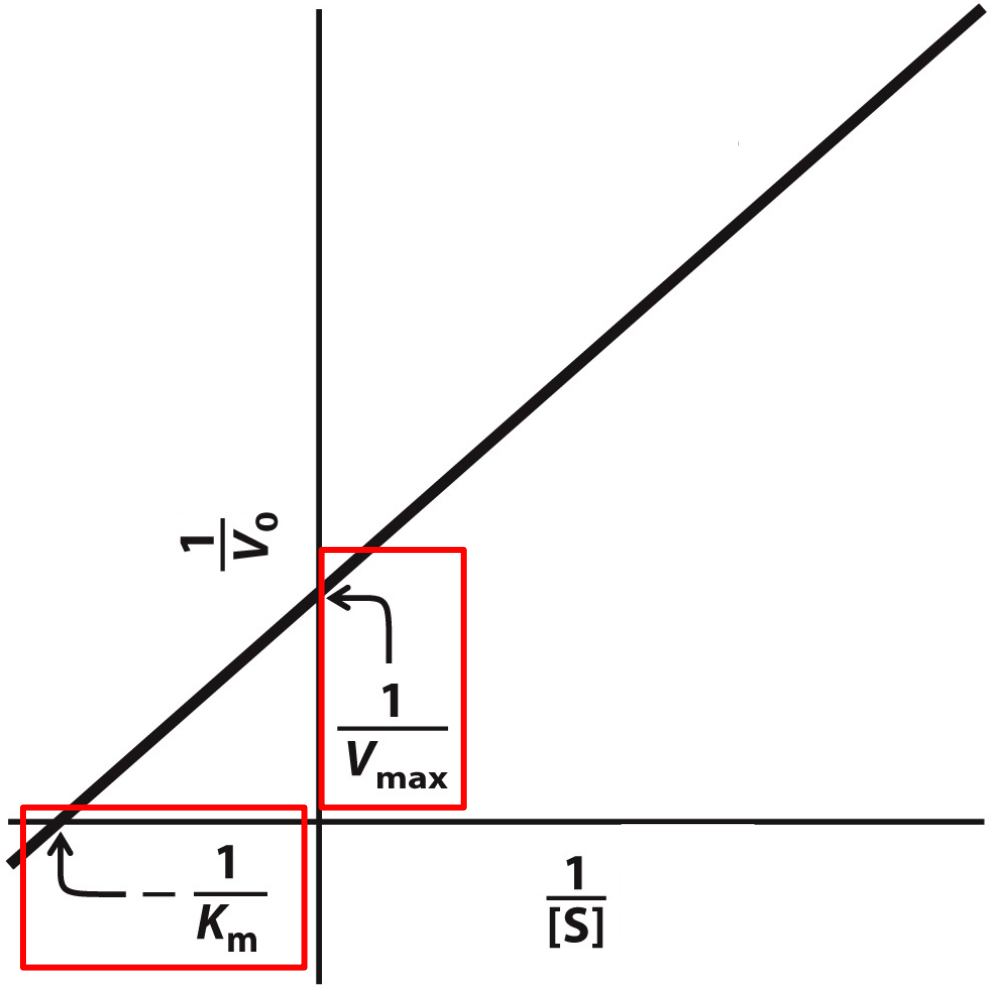
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Graphical determination of kinetic parameters

Lineweaver-Burk (double-reciprocal) plot

$$\frac{1}{V_0} = \frac{K_m}{V_{max}} \cdot \frac{1}{[S]} + \frac{1}{V_{max}}$$

$y = ax + b$

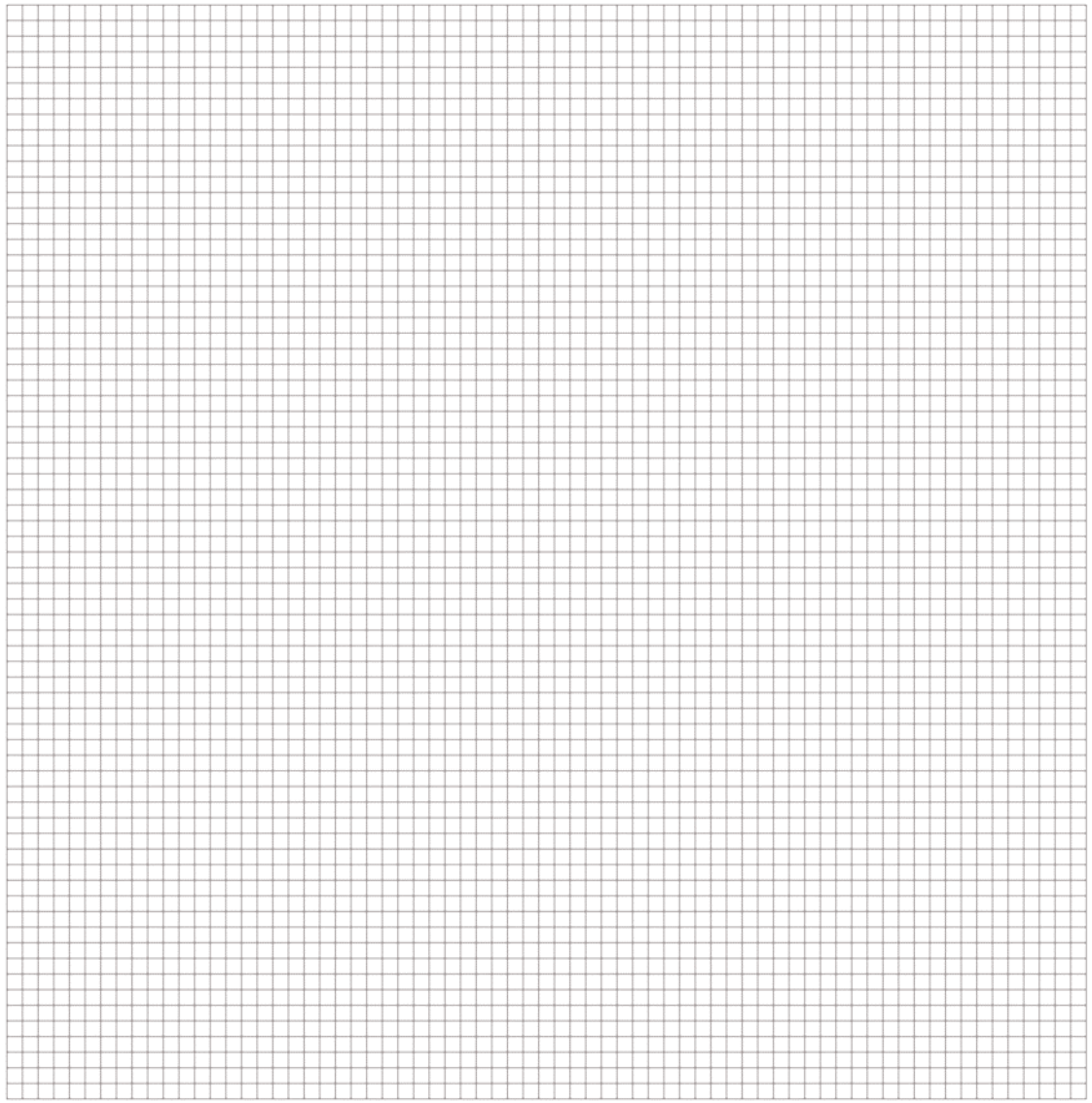


Box 6-1 Figure 1
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[E] = 1 µM

[S] (mM)	V_0 (mM/s)	1/[S]	$1/V_0$
0.2	0.04		
1	0.13		
2	0.19		
4	0.24		
20	0.31		

K_m (mM)	
V_{max} (mM/s)	
k_{cat} (s^{-1})	
K_{cat}/k_m ($mM^{-1}s^{-1}$)	



Enzyme inhibitors

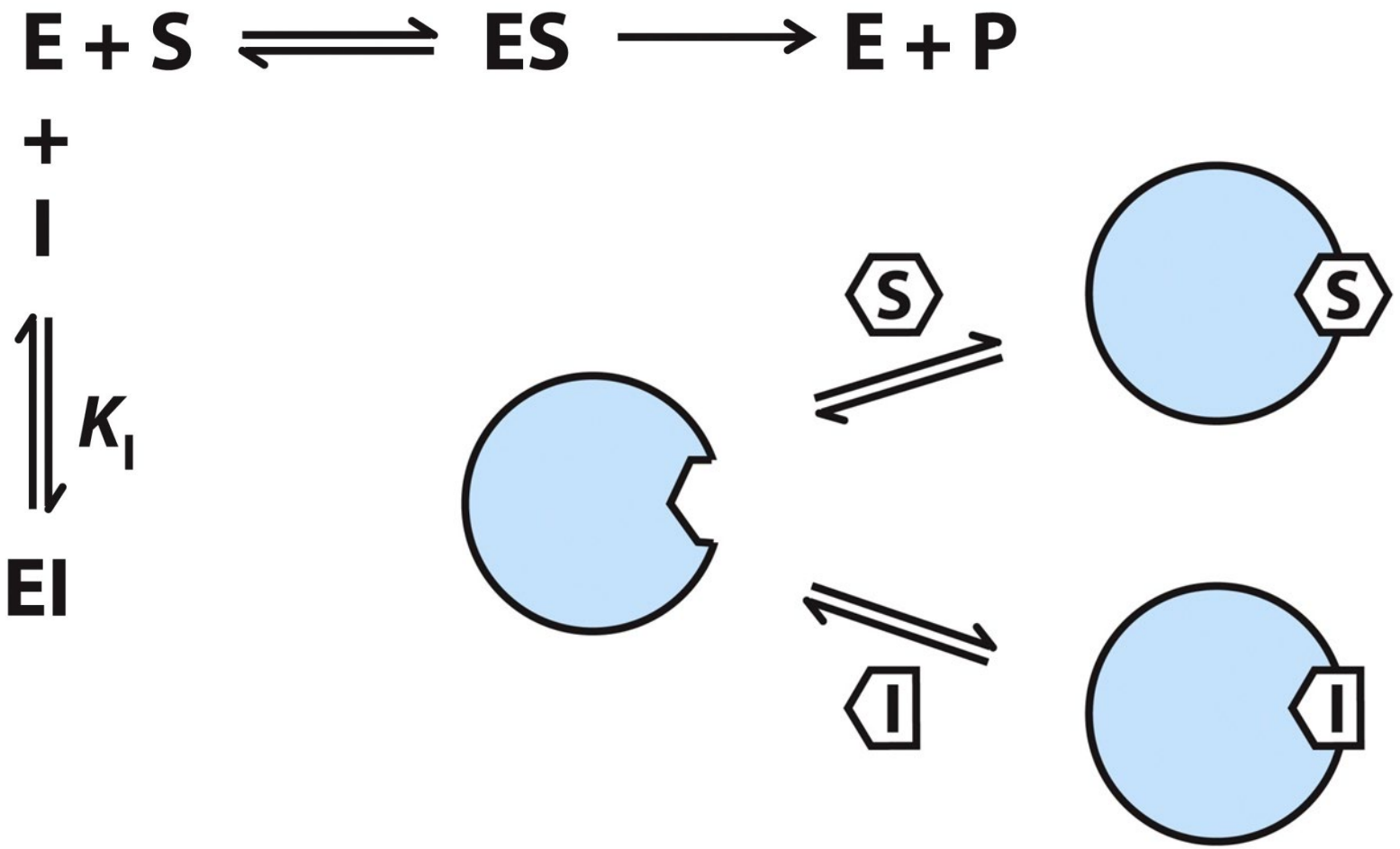
Inhibitors are compounds that decrease enzyme activity

- Irreversible inhibitors permanently inactivate the enzyme
  - They are often powerful toxins, but also may be used as drugs
- Reversible inhibitors bind to and can dissociate from the enzyme
  - They are often used as drugs to slow down a specific enzyme
- Inhibitors can be complete or partial
- Inhibitors can bind:
  - to the free enzyme and prevent the binding of the substrate
  - to the enzyme-substrate complex and prevent the reaction

Types of reversible inhibition

- Competitive inhibition
- Uncompetitive inhibition
- Non-competitive inhibition

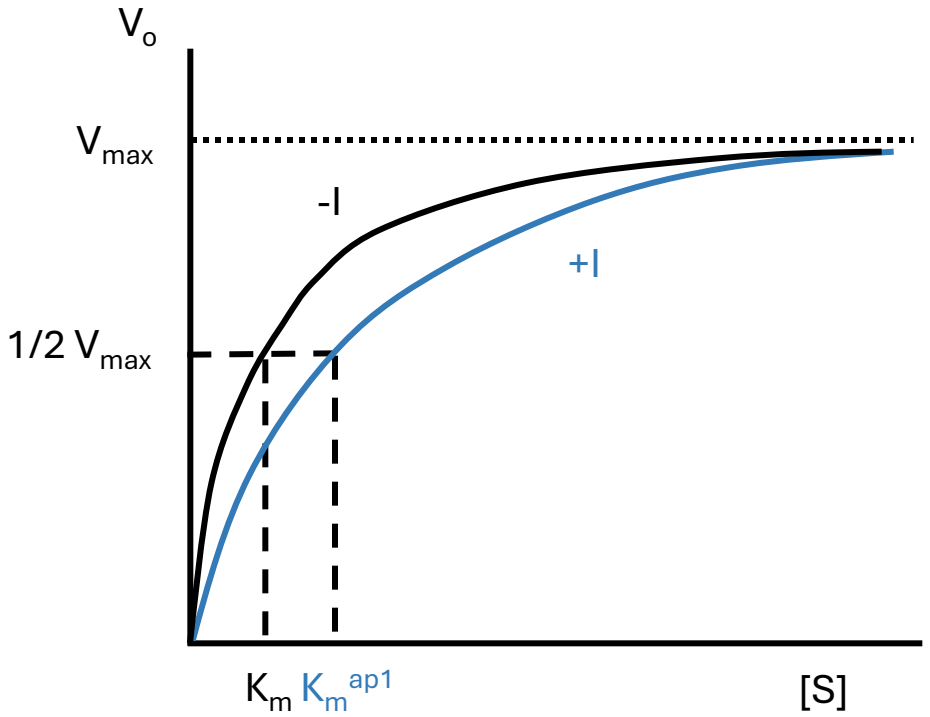
Competitive inhibition

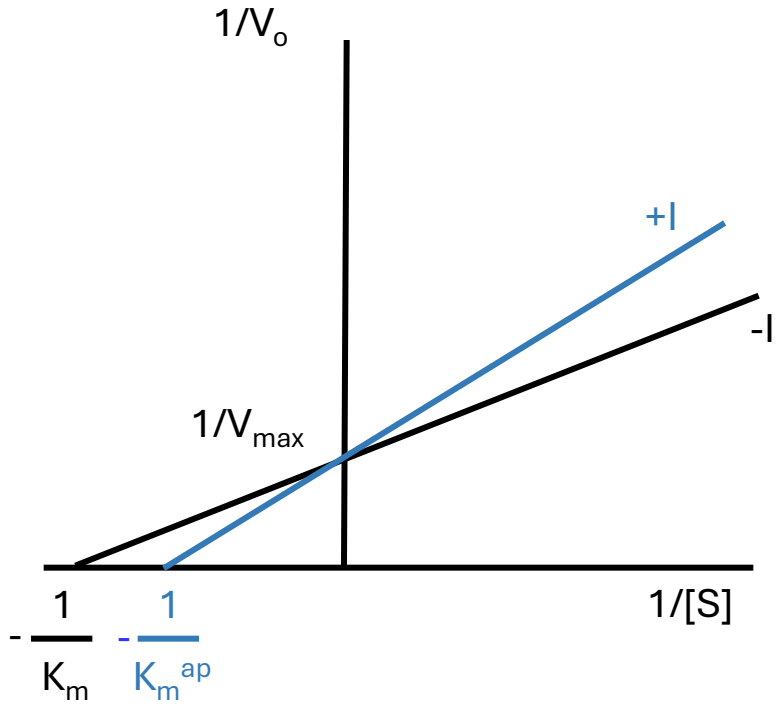


The inhibitor binds to the active site, competing with the substrate

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Competitive inhibition

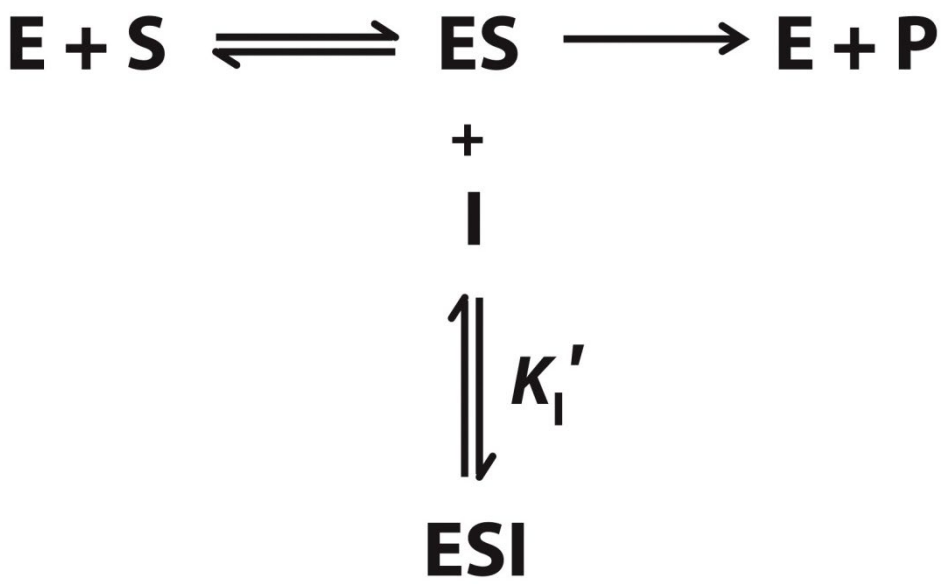


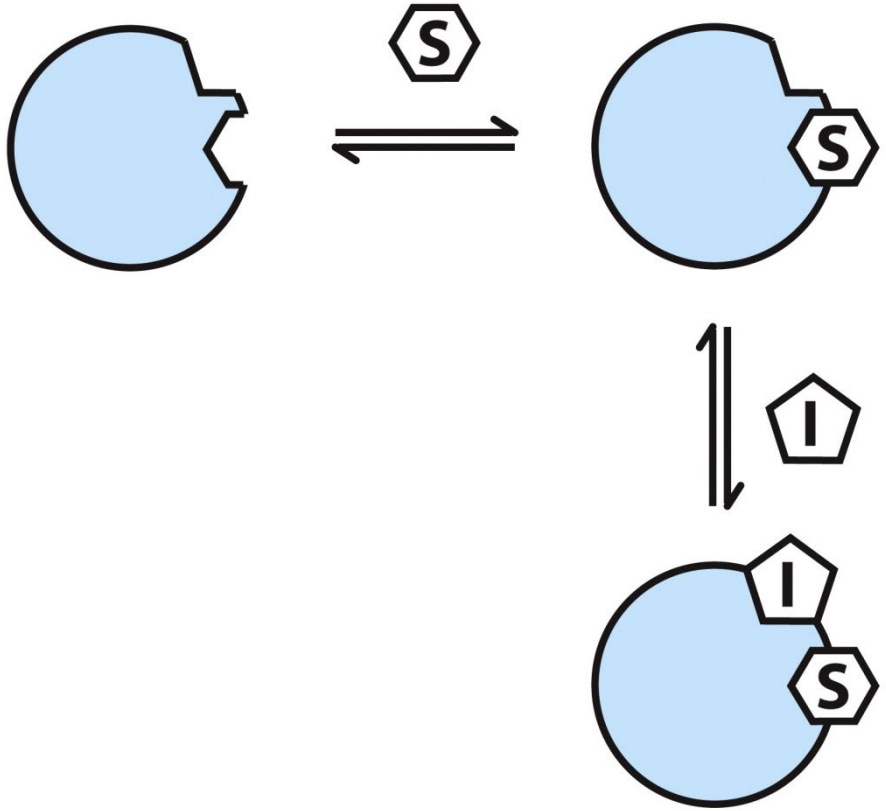


Increase in K_m

No change in V_{max}

Uncompetitive inhibition

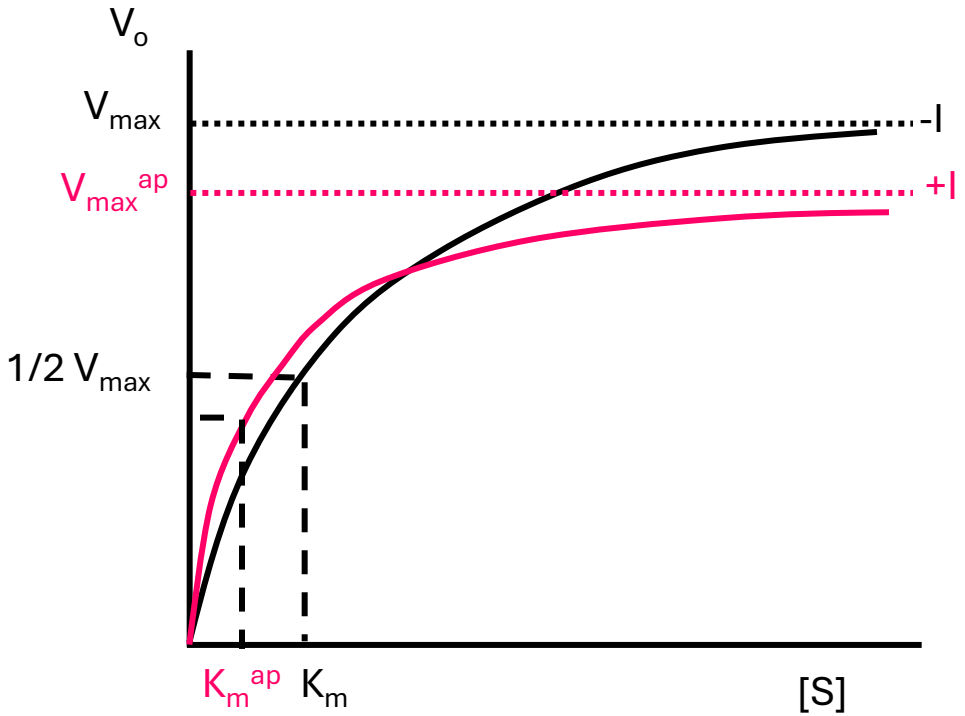


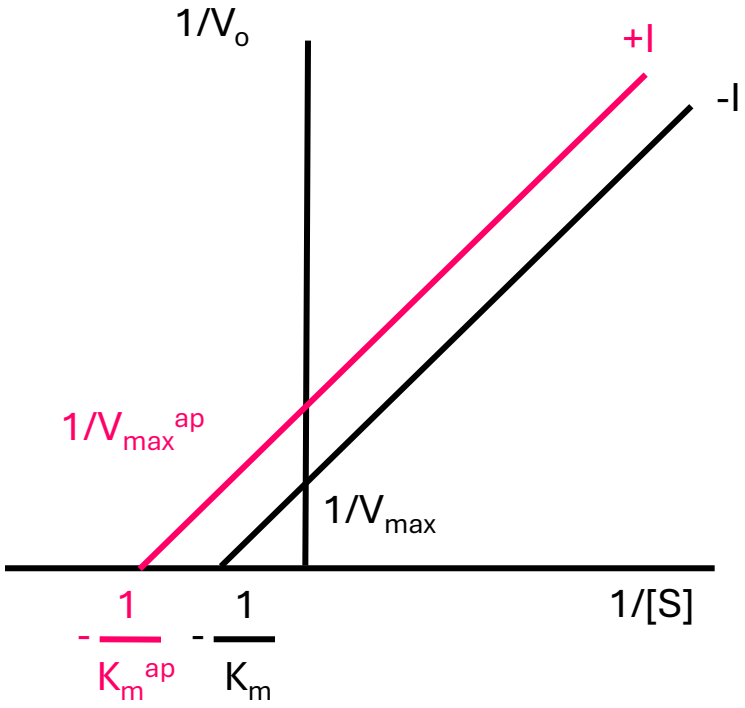


The inhibitor does not affect substrate binding, but inhibits the catalytic function by binding to the enzyme-substrate complex

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Uncompetitive inhibition



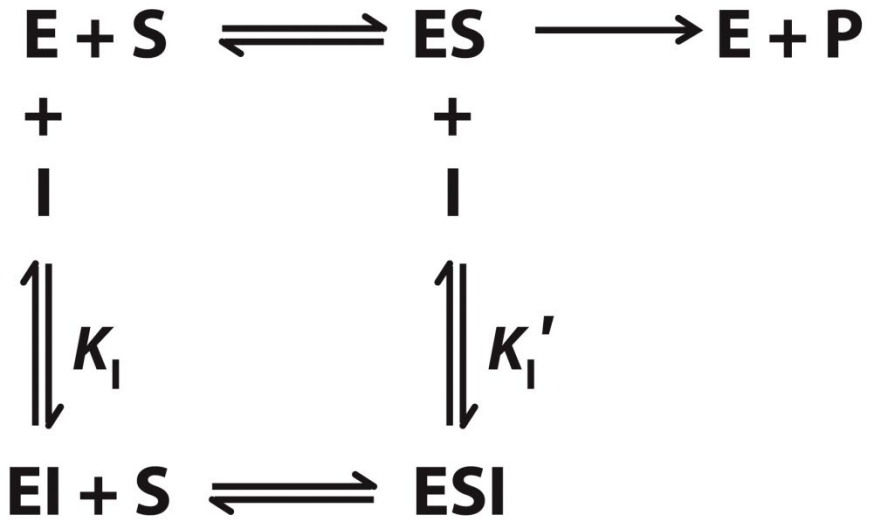


Decrease in K_m

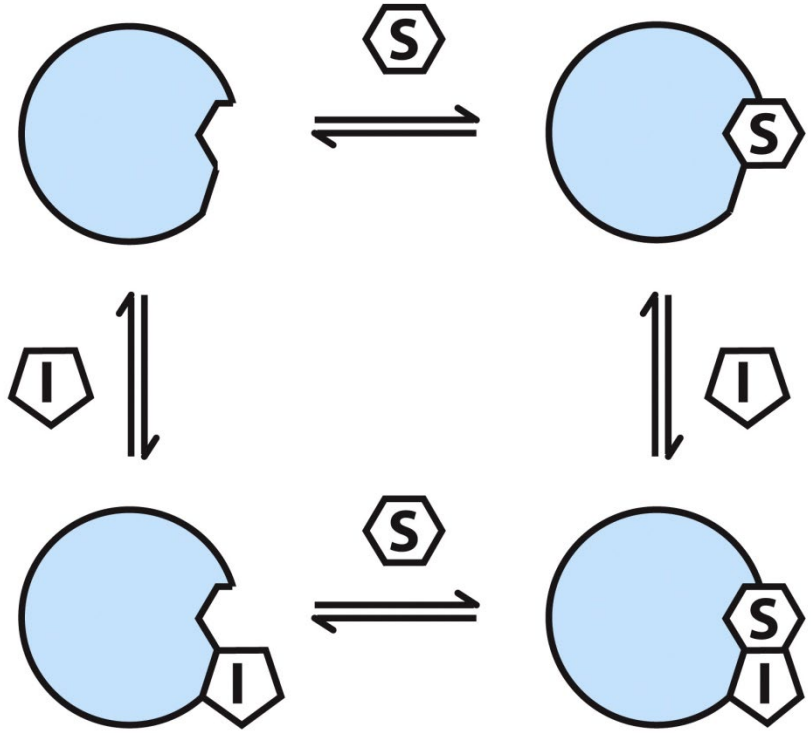
Decrease in V_{max}

No change in K_m/V_{max}

Noncompetitive inhibition



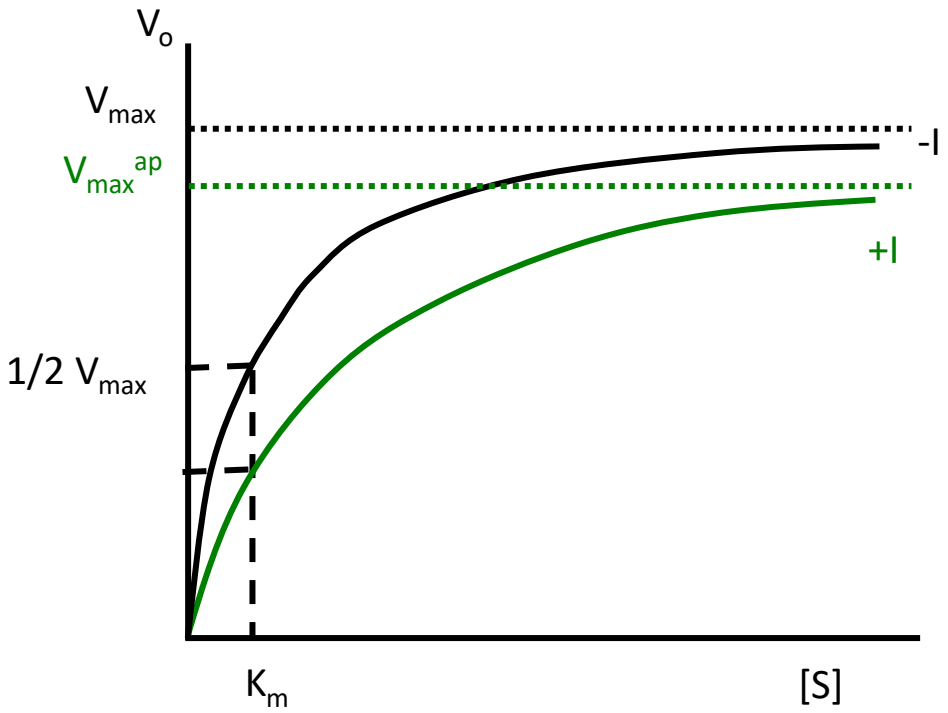
$$K_I = K_I'$$

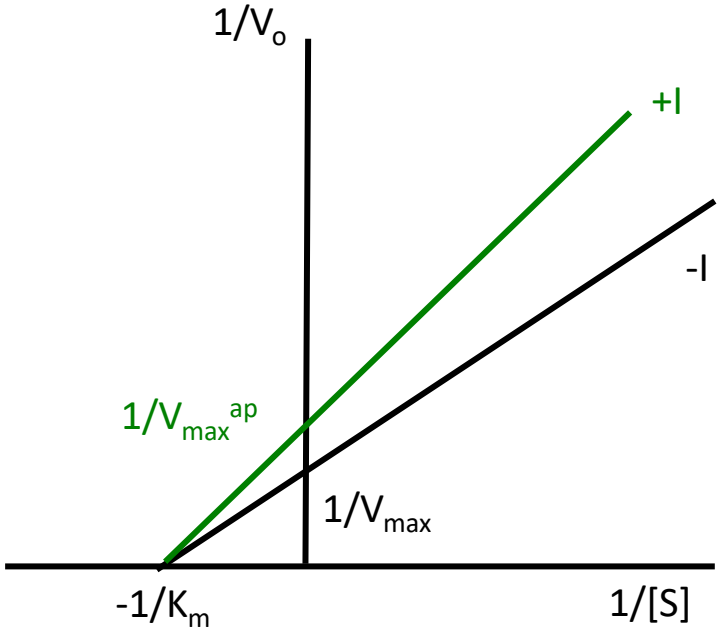


The inhibitor binds both to the free enzyme and to the enzyme-substrate complex

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Noncompetitive inhibition





No change in K_m

Decrease in V_{max}

[E] = 1 µM

+ Inhibitor A (10 nM)

[S] (mM)	V_0 (mM/s)	1/[S]	$1/V_0$		
0.2	0.02			K_m (mM)	
1	0.07			V_{max} (mM/s)	
2	0.09			k_{cat} (s^{-1})	
4	0.12			K_{cat}/k_m ($mM^{-1}s^{-1}$)	
20	0.18			Inhibition type	

+ Inhibitor B (10 nM)

[S] (mM)	V_0 (mM/s)	1/[S]	$1/V_0$		
0.2	0.03			K_m (mM)	
1	0.07			V_{max} (mM/s)	
2	0.08			k_{cat} (s^{-1})	
4	0.09			K_{cat}/k_m ($mM^{-1}s^{-1}$)	
20	0.1			Inhibition type	

+ Inhibitor C (10 nM)

[S] (mM)	V_0 (mM/s)	1/[S]	$1/V_0$		
0.2	0.02			K_m (mM)	
1	0.08			V_{max} (mM/s)	
2	0.14			k_{cat} (s^{-1})	
4	0.2			K_{cat}/k_m ($mM^{-1}s^{-1}$)	
20	0.25			Inhibition type	